

MBTA GTFS Documentation

About this document

This document describes the MBTA's implementation of the General Transit Feed Specification (GTFS). It is intended to supplement, not replace, [documentation on GTFS itself](#).

The MBTA GTFS feed is available from the following URL: https://cdn.mbta.com/MBTA_GTFS.zip.

About the fields in this document

- **Status** field
 - **Included**
 - **Included (some records)**
 - **Included (empty)**: Included, but not populated
 - **N/A**: Part of GTFS specifications, but not included in MBTA GTFS
 - **Coming soon**: Will be included in upcoming GTFS version
 - **Deprecated**: No longer supported
 - **Persistent**: Value for record will not change between one **feed_version** and the next
- **GTFS spec** field
 - **Required**
 - **Optional**
 - **Experimental**: Not part of GTFS specifications, but included in MBTA GTFS

GTFS Files

Table Name	GTFS spec	Status	Notes
agency.txt	Required	Included	
areas.txt	Optional	Included	Provides labels for groupings of locations further defined in stop_areas.txt .
attributions.txt	Optional	N/A	
calendar.txt	Required	Included	Generated programatically. May not be easy for humans to read. May include <code>service_ids</code> used exclusively in timeframes.txt , rather than trips.txt .
calendar_attributes.txt	Experimental	Included	Adds human-readable names and further context to calendar <code>service_ids</code> .
calendar_dates.txt	Optional	Included	Generated programatically. May not be easy for humans to read. May include <code>service_ids</code> used exclusively in timeframes.txt , rather than trips.txt .
checkpoints.txt	Experimental	Included	Similar in part to stops.txt , this table provides human-readable names to checkpoints from stop_times.txt .
directions.txt	Experimental	Included	Provides for passenger-facing names to be documented for <code>direction_id</code> on a route-by-route basis.
facilities.txt	Experimental	Included	Amenities such as elevators, escalators, parking lots, and bike storage. Typically located within stations.
facilities_properties.txt	Experimental	Included	Properties of station amenities in facilities.txt .
facilities_properties_definitions.txt	Experimental	Included	Definitions of the <code>property_ids</code> and values used in facilities_properties.txt .
fare_attributes.txt	Optional	N/A	
fare_leg_join_rules.txt	Optional	Included	Rules for treating multiple fare legs as a single effective fare leg.
fare_leg_rules.txt	Optional	Included	Rules for calculating fares for individual legs of travel. Does not include all possible MBTA fares.
fare_media.txt	Optional	Included	Describes selected media that can be utilized to use fare products.
fare_products.txt	Optional	Included	Describes different types of fares that can be purchased by riders. Does not include all possible MBTA fares.
fare_rules.txt	Optional	N/A	
fare_transfer_rules.txt	Optional	Included	Rules for calculating fares for transfers between legs of travel.
feed_info.txt	Optional	Included	
frequencies.txt	Optional	N/A	
levels.txt	Optional	Included	Provides relative elevation information for <code>stop_ids</code> (including boarding platforms, station entrances, and generic nodes) within a parent station.
lines.txt	Experimental	Included	Groups similar routes (such as those which serve the same trunk corridor or bus terminal) for the purpose of customer display.
linked_datasets.txt	Experimental	Included	URLs to linked GTFS-realtime datasets: Trip Updates, Vehicle Positions and Service Alerts.

Table Name	GTFS spec	Status	Notes
multi_route_trips.txt	Experimental	Included	For trips that travel on more than one route, this file identifies additional routes with which the trip should be associated.
pathways.txt	Optional	Included	Information, including travel times and lengths, about paths within and out of parent stations, including paths between platforms and to/from station entrances.
routes.txt	Required	Included	
route_patterns.txt	Experimental	Included	Information about the different variations of service that may be run within a single route_id, including when and how often they are operated.
shapes.txt	Optional	Included	
stops.txt	Required	Included	
stop_areas.txt	Optional	Included	Assigns selected stops to area groupings for the purpose of calculating fares.
stop_times.txt	Required	Included	
timeframes.txt	Optional	Included	Date and time periods to use in fare rules for fares that vary based on date and time factors. Learn more about the recent introduction of timeframes to GTFS.
transfers.txt	Optional	Included	
translations.txt	Optional	N/A	
trips.txt	Required	Included	
trips_properties.txt	Experimental	Included	Identifies special properties and notes relating to trips appearing in trips.txt .
trips_properties_definitions.txt	Experimental	Included	Definitions of the trip_property_ids and values used in trips_properties.txt .

agency.txt

Agencies in the MBTA GTFS feed include the MBTA itself and [Cape Cod RTA](#) (for the CapeFLYER). The CapeFLYER may move to a separate GTFS feed in the future, in which case CCRTA would be removed from this file.

Field Name	GTFS spec	Status	Notes
agency_id	Optional	Included	MBTA agency_id = 1.
agency_name	Required	Included	
agency_url	Required	Included	
agency_timezone	Required	Included	
agency_lang	Optional	Included	
agency_phone	Optional	Included	
agency_fare_url	Optional	Included	
agency_email	Optional	N/A	

areas.txt

Field Name	GTFS spec	Status	Notes
area_id	Required	Included	
area_name	Optional	Included	

calendar.txt

Services defined in this file may be referenced in either [trips.txt](#) or [timeframes.txt](#).

Field Name	GTFS spec	Status	Notes
service_id	Required	Included	
monday	Required	Included	
tuesday	Required	Included	
wednesday	Required	Included	
thursday	Required	Included	
friday	Required	Included	
saturday	Required	Included	
sunday	Required	Included	
start_date	Required	Included	
end_date	Required	Included	

calendar_attributes.txt

Adds human-readable names to calendar service_ids and further information about when they operate and how closely the service aligns to service on a typical day. This specification adds onto the [GTFS+ calendar_attributes.txt extension](#).

Field Name	GTFS spec	Status	Notes
service_id	Experimental	Included	Linked to values in calendar.txt and calendar_dates.txt .
service_description	Experimental	Included	Human-readable description of the service, as it should appear on public-facing websites and applications. Some of the possible values include, but are not limited to, "Weekday schedule", "Weekday schedule (no school)", "Saturday schedule", "Saturday schedule (modified)", "Holiday (Sunday schedule)", and "Storm schedule (reduced service)".
service_schedule_name	Experimental	Included	Description of when the service_id is in effect. Typically will look like "Weekday", "Weekday (no school)", "Holiday", "Saturday", "Sunday", and "Storm (reduced schedule)".
service_schedule_type	Experimental	Included	Description of the schedule type the service_id can be applied. For example, on a holiday, the service_schedule_type value may be "Saturday" or "Sunday". Current valid values are: - Weekday - Weekend - Saturday - Sunday - Other
service_schedule_typicality	Experimental	Included	Describes how well this schedule represents typical service for the listed service_schedule_type. Current valid values are: 0 (or empty): Not defined. 1: Typical service with perhaps minor modifications. 2: Extra service supplements typical schedules. 3: Reduced holiday service is provided by typical Saturday or Sunday schedule. 4: Major changes in service due to a planned disruption, such as construction. 5: Major reductions in service for weather events or other atypical situations. 6: Not actively scheduled. Used to hold reference trips showing a route's canonical set of stops. The stop times for such trips are not representative of actual service.
rating_start_date	Experimental	Included	Start date of the schedule rating of which this service is a part. It will be earlier than or equal to the service's start_date.
rating_end_date	Experimental	Included (some records)	End date of the schedule rating of which this service is a part. It will be later than or equal to the service's end_date. This value may be empty, in which case the schedule rating is effective from the rating_start_date until further notice.
rating_description	Experimental	Included	Human-readable name of this schedule rating. Typically will be "Winter", "Spring", "Summer", or "Fall".

calendar_dates.txt

Services defined in this file may be referenced in either [trips.txt](#) or [timeframes.txt](#).

Field Name	GTFS spec	Status	Notes
service_id	Required	Included	
date	Required	Included	
exception_type	Required	Included	
holiday_name	Experimental	Included (some records)	Names a particular holiday for which the service exception is in effect (e.g. "Thanksgiving Day").

checkpoints.txt

This table provides human-readable names to checkpoints from [stop_times.txt](#). The tables are joined on the experimental checkpoint_id field.

Field Name	GTFS spec	Status	Notes
checkpoint_id	Experimental	Included	Values in this field must be unique.
checkpoint_name	Experimental	Included	A general location can have multiple checkpoints; values in this field are not necessarily unique.

directions.txt

Provides for passenger-facing names to be documented for each direction_id value on a route-by-route basis. This specification adds onto the [GTFS+ directions.txt extension](#).

Field Name	GTFS spec	Status	Notes
route_id	Experimental	Included	Linked to routes.txt file.

Field Name	GTFS spec	Status	Notes
direction_id	Experimental	Included	Binary value linked to trips.txt file. Combining route_id and direction_id forms the unique primary key for this file.
direction	Experimental	Included	Human-readable text corresponding to a direction name in a given list. Current valid values are: • North • South • East • West • Northeast • Northwest • Southeast • Southwest • Clockwise • Counterclockwise • Inbound • Outbound • Loop A • Loop B • Loop
direction_destination	Experimental	Included	MBTA extension which provides the customer-facing direction name used on in-station wayfinding. It can, but need not, include all possible destinations; it could instead list a cardinal direction or city/town. For example, direction_id 0 on route_id Red may have a direction value of South, and a direction_destination of Ashmont/Braintree.

facilities.txt

Field Name	GTFS spec	Status	Notes
facility_id	Experimental	Included	Uniquely identifies each facility. Must be unique within this table.
facility_code	Experimental	Included (some records)	Number or code to identify facility to customers. If the facility is labeled with a number in practice, that is used for a code.
facility_class	Experimental	Included	Specifies overarching category of a facility. Current valid values are: • 0: Unidentified or other • 1: Pedestrian access (Examples include: walkway, elevator, stairway) • 2: Wheelchair access (Exclusively to facilitate wheelchair movement, with examples such as: wheelchair lift, bridge plate, ramp) • 3: Private mode access (Examples include: parking garage, parking space, bike storage, pick-up/drop-off, taxi stand) • 4: Fare collection (Examples include: fare vending machine, turnstile array, turnstile, ticket window, fare collection store, retail sales location) • 5: Information and assistance (Examples include: customer service desk, customer assistance phone, emergency phone, schedule card rack, countdown sign) • 6: Amenities (Examples include: wi-fi, payphone, bench, artwork, waiting area)
facility_type	Experimental	Included	Specifies the direct type of a facility, answering the question, "What is this facility?" Current valid values are: • elevator • escalator • ramp • elevated-subplatform (also called a "mini-high") • fully-elevated-platform • portable-boarding-lift • bridge-plate • parking-area • pick-drop • taxi-stand • bike-storage • electric-car-chargers • fare-vending-machine • fare-vending-retailer • fare-media-assistant • fare-media-assistance-facility • ticket-window • other
stop_id	Experimental	Included (some records)	Specifies one stop_id with which the facility is associated. Common field with stops.txt . If a facility is associated with more than one stop_id within a parent station, the stop_id of the parent station should fill this field. In this case, if the facility does not serve every child stop within the parent stop, each excluded child stop served must be specified as a separate entry in facilities_properties.txt with the property excludes-stop.
facility_short_name	Experimental	Included (some)	Contains a short name of the facility. Note that this name might not include the station name or the type of facility. At least one of facility_short_name or facility_long_name must

Field Name	GTFS spec	Status	Notes
		records)	be specified, or potentially both if appropriate.
facility_long_name	Experimental	Included (some records)	Contains a descriptive name of the facility which can be used without additional context concerning the station or type of facility. At least one of facility_short_name or facility_long_name must be specified, or potentially both if appropriate.
facility_desc	Experimental	Included (some records)	
facility_lat	Experimental	Included (some records)	
facility_lon	Experimental	Included (some records)	
wheelchair_facility	Experimental	Included	Identifies whether a facility is accessible to customers using a wheelchair. The field can have the following values: 0 (or empty): indicates that there is no accessibility information for the facility 1: indicates that the facility is wheelchair accessible 2: facility not accessible to persons in wheelchairs

facilities_properties.txt

Field Name	GTFS spec	Status	Notes
facility_id	Experimental	Included	Uniquely identifies each facility as defined in facilities.txt . Does not need to be unique within this table.
property_id	Experimental	Included	Property or function for which details are provided in the value field. This may describe a particular function not described by facility_class in facilities.txt , or another property describing a detail about the facility.
value	Experimental	Included	Value corresponding to a particular property for a facility_id.

facilities_properties_definitions.txt

Field Name	GTFS spec	Status	Notes
property_id	Experimental	Included	Property ID which is referenced from the property field in facilities_properties.txt .
definition	Experimental	Included	Text describing the associated property_id, its intended usage.
possible_values	Experimental	Included	Text describing the allowed values and format of the property_id's respective value field.

fare_leg_rules.txt

Currently describes fare leg rules for full-fare passengers paying walk-up fares. Fares for passengers eligible for reduced fares, as well as those involving the use of passes and stored-value CharlieCards, to be introduced to the MBTA's GTFS at a later date.

More complete information about the MBTA's fares, transfers, and passes can be found at [MBTA.com/fares](https://www.mbta.com/fares). Additionally, [MBTA.com/tariff](https://www.mbta.com/tariff) provides a technical reference for the MBTA's fare policy implementation.

Field Name	GTFS spec	Status	Notes
leg_group_id	Optional	Included	
network_id	Optional	Included	
from_area_id	Optional	Included	
to_area_id	Optional	Included	
fare_product_id	Required	Included	
from_timeframe_group_id	Optional	Included (some records)	
to_timeframe_group_id	Optional	Included (some records)	
transfer_only	Experimental	Included	This is a proposed Transit extension to designate fare rules that may only be considered as as part of a transfer. Consumers of the MBTA's fares data are strongly encouraged to integrate this field to ensure accurate rapid transit fares are calculated. Valid values: 0 (or empty): This rule may be considered regardless of whether any transfer rules apply 1: This rule may only be considered if there is a matching fare transfer rule from the preceding leg to the current leg; a rule with this value can never

Field Name	GTFS spec	Status	Notes
			match the first leg of a journey

fare_media.txt

Details the physical and virtual fare media that can be used for the fares described by [fare_products.txt](#) and [fare_leg_rules.txt](#). CharlieCards to be introduced to this file at a later date.

Field Name	GTFS spec	Status	Notes
fare_media_id	Required	Included	
fare_media_name	Optional	Included	
fare_media_type	Required	Included	A 1 value represents a physical, paper ticket that allows a passenger to take either a certain number of pre-purchased trips or unlimited trips within a fixed period of time.

fare_products.txt

Promotional fares and unlimited passes may be introduced to this file at a later date. More complete information about the MBTA's fare and pass products can be found at MBTA.com/fares.

Field Name	GTFS spec	Status	Notes
fare_product_id	Required	Included	
fare_product_name	Optional	Included	
fare_media_id	Optional	Included	
amount	Required	Included	
currency	Required	Included	

fare_transfer_rules.txt

Only those transfers permitted with CharlieTickets are currently included in this file.

More complete information about the MBTA's allowed transfers, including those not yet found in GTFS, can be found at MBTA.com/transfers.

Field Name	GTFS spec	Status	Notes
from_leg_group_id	Optional	Included	
to_leg_group_id	Optional	Included	
transfer_count	Required	Included	
duration_limit	Optional	Included	
duration_limit_type	Required	Included	
fare_transfer_type	Required	Included	
fare_product_id	Optional	Included	
filter_fare_product_id	Experimental	Included	Identifies the fare product that must be used in both the pre-transfer leg identified by from_leg_group_id and the post-transfer leg identified by to_leg_group_id, such as a pass. If empty, this transfer rule applies to all fare products that match in both legs. Conditionally Forbidden if fare_transfer_rules.fare_product_behavior has a value of 0 or is empty.
fare_media_behavior	Experimental	Included	Identifies whether the rule applies only when the same fare media is used in both the pre-transfer leg identified by from_leg_group_id and the post-transfer leg identified by to_leg_group_id. Valid options are: 0 or empty: The transfer rule only applies when both legs use the same fare media. 1: The transfer rule applies regardless of which media are used in both legs.
fare_product_behavior	Experimental	Included	Identifies whether the rule applies only when the same fare product is used in both the pre-transfer leg identified by from_leg_group_id and the post-transfer leg identified by to_leg_group_id. Valid options are: 0 or empty: The transfer rule applies regardless of which products are used in both legs. 1: The transfer rule only applies when both legs use the same fare product.

fare_leg_join_rules.txt

Currently only applies to Commuter Rail fare legs transferring at stations with a high number of coordinated transfers between trips. May be expanded in the future to cover additional stations and/or modes.

Field Name	GTFS spec	Status	Notes
from_network_id	Required	Included	
to_network_id	Required	Included	
from_stop_id	Conditionally Required	Included	
to_stop_id	Conditionally Required	Included	

feed_info.txt

Field Name	GTFS spec	Status	Notes
feed_publisher_name	Required	Included	
feed_publisher_url	Required	Included	
feed_lang	Required	Included	
feed_start_date	Optional	Included	When comparing two GTFS feeds, the newer one will always have a newer feed_start_date.
feed_end_date	Optional	Included	
feed_version	Optional	Included	
feed_contact_email	Optional	Included	An email address for communication regarding the GTFS dataset and data publishing practices. It is a technical contact for GTFS-consuming applications. Provide customer service contact information through agency.txt .
feed_id	Experimental	Included	The universally unique identifier for the GTFS feed.

levels.txt

Used to describe the vertical levels within a station. level_ids can be applied to stops in [stops.txt](#). [Learn more about levels in GTFS.](#)

Field Name	GTFS spec	Status	Notes
level_id	Required	Included	The level_id field contains an ID that uniquely identifies the level. The level_id is dataset unique. Reuse of level_ids within different stations does not imply that the stations are connected together.
level_index	Required	Included	Relative position of a level within a single station. In general, a value of 0 indicates the ground (or street) level, position values indicate levels above ground, and negative values indicate levels below ground. For some stations on slopes, it may be necessary to have multiple levels to represent adjacent streets, if the streets are at different elevations.
level_name	Optional	Included	

lines.txt

In this file, a line is a combination of existing routes in [routes.txt](#). This concept can be used to group similar routes when displaying them to customers, such as for routes which serve the same trunk corridor or bus terminal. Lines have similar attributes as routes, such as short and long names, descriptions, URLs, colors, and sort ordering.

Field Name	GTFS spec	Status	Notes
line_id	Experimental	Included	Feed-unique identifier for lines, or groups of routes. This ID will be referenced in routes.txt for routes which are contained in this line.
line_short_name	Experimental	Included	Short, public-facing name for the group of routes represented in this line. Usage between this field and line_long_name is the same as the relationship between route_short_name and route_long_name.
line_long_name	Experimental	Included	Lengthier, public-facing name for the group of routes represented in this line. Usage between this field and line_short_name is the same as the relationship between route_short_name and route_long_name.
line_desc	Experimental (empty)	Included	Contains a human-readable description of the line.
line_url	Experimental	Included (empty)	Contains the URL of a web page about the particular line. This should be different from the agency_url or any route_url values. The value must be a fully qualified URL that includes http:// or https://, and any special characters in the URL must be correctly escaped.
line_color	Experimental	Included	In systems that have colors assigned to lines, the route_color field defines a color that corresponds to a line. The color must be provided as a six-character hexadecimal number, for example, 00FFFF. If no color is specified, the default route color is white (FFFFFF).
line_text_color	Experimental	Included	This field can be used to specify a legible color to use for text drawn against a background of line_color. The color must be provided as a six-character hexadecimal number, for example, FFD700. If no color is specified, the default text color is black (000000).

Field Name	GTFS spec	Status	Notes
line_sort_order	Experimental	Included	The line_sort_order field can be used to order the lines in a way which is ideal for presentation to customers. It must be a non-negative integer. Lines with smaller line_sort_order values should be displayed before lines with larger line_sort_order values.

linked_datasets.txt

Experimental file containing URLs and authentication information for linked GTFS-realtime datasets: Trip Updates, Vehicle Positions and Service Alerts. [Learn more about the proposed linked_datasets extension to GTFS.](#)

Field Name	GTFS spec	Status	Notes
url	Experimental	Included	Contains the URL to the linked dataset.
trip_updates	Experimental	Included	Indicates whether the dataset at the specified URL may contain a TripUpdate entity . Valid values are 0 if the dataset does not contain TripUpdate entities, or 1 if the dataset may contain TripUpdate entities.
vehicle_positions	Experimental	Included	Indicates whether the dataset at the specified URL may contain a VehiclePosition entity . Valid values are 0 if the dataset does not contain VehiclePosition entities, or 1 if the dataset may contain VehiclePosition entities.
service_alerts	Experimental	Included	Indicates whether the dataset at the specified URL may contain an Alert entity . Valid values are 0 if the dataset does not contain Alert entities, or 1 if the dataset may contain Alert entities.
authentication_type	Experimental	Included	Defines the type of authentication required to access the URL. Presently, all included URLs have a value of 0 indicating that no authentication is required.

multi_route_trips.txt

Some transit trips serve more than one route. For example, Commuter Rail trips that travel partly on the Lowell Line and partly on the Haverhill Line, or bus trips labeled "Route 62/76" that should be shown on the respective schedules for both Route 62 and Route 76. This applies when a trip serves one route and then the other, but also when the trip serves a unique combination of stops from each route.

This experimental file indicates routes that a trip is associated with, in addition to the route_id identified with this trip in [trips.txt](#). If showing a schedule of all service on a route, that display should include trips associated with the route in [multi_route_trips.txt](#). If showing general information about a trip in a different context, the route_id identified with the in [trips.txt](#) is still the best route to label the trip with.

If every trip of a route_id appears here, that route does not have any trips of "its own" and should probably not be shown to customers in lists of routes.

Field Name	GTFS spec	Status	Notes
added_route_id	Experimental	Included	An additional route_id associated with the given trip_id. Can appear repeatedly if more than one trip is added to it.
trip_id	Experimental	Included	The trip_id of the trip which has the additional route associated with it. Can appear repeatedly if it is being added to more than one route.

pathways.txt

Describes the various pedestrian paths of travel within and around a station. [Learn more about pathways in GTFS.](#)

At this time, the MBTA implementation of pathways does not include all stations, with additional stations being added to the file continually.

Field Name	GTFS spec	Status	Notes
pathway_id	Required	Included	The pathway_id field contains an ID that uniquely identifies the pathway. The pathway_id is dataset unique.
from_stop_id	Required	Included	Stop IDs are referenced from the stops.txt file, and include service stops, station entrances, and generic nodes.
to_stop_id	Required	Included	Stop IDs are referenced from the stops.txt file, and include service stops, station entrances, and generic nodes.
facility_id	Experimental	Included (some records)	Links pathways to facilities (in facilities.txt) which may be required to be transversed, such as escalators and elevators. Can be coupled with real-time alerts feeds/API calls to determine if the pathway is available at a given time.
pathway_mode	Required	Included	Type of pathway. Values used by the MBTA for this field are: 1: Walkways, including ramps 2: Staircases

Field Name	GTFS spec	Status	Notes
			4: Escalators 5: Elevators 6: Entering fare control (by passing through fare gates) 7: Exiting fare control (by passing through fare gates or exit-only gates)
is_bidirectional	Required	Included	For the MBTA implementation, this field will always be 0. All pathways will be unidirectional.
length	Optional	Included (some records)	Length is provided in meters.
wheelchair_length	Experimental	Included (some records)	This field is temporarily added for backward compatibility alongside wheelchair_traversal_time and will be deprecated and removed in the future. Wheelchair length is provided in meters.
traversal_time	Optional	Included (some records)	Traversal time is provided in seconds.
wheelchair_traversal_time	Experimental	Included (some records)	This field is included for backward compatibility and will be deprecated and removed in the future. Wheelchair traversal time is provided in seconds.
stair_count	Optional	Included (some records)	Number of stairs required to traverse, in positive integers if stairs are to be ascended, and in negative integers if stairs are to be descended.
max_slope	Optional	Included (some records)	Will be populated and non-zero for sloped pathways.
pathway_name	Experimental	Included	Contains description of the path origin and destination.
pathway_code	Experimental	Included (empty)	
signposted_as	Optional	Included (some records)	Contains indication of the path destination as it is signed at the particular station. If no such signage exists, the field will be blank.
instructions	Experimental	Included (empty)	

routes.txt

Field Name	GTFS spec	Status	Notes
route_id	Required	Included (persistent)	
agency_id	Optional	Included (persistent)	
route_short_name	Required	Included (some records)	Populated for service with branches on separate route IDs (such as Green Line or Silver Line) and all MBTA bus services. If route_short_name is populated for a route, regardless of the presence of route_long_name, then "Route route_short_name" is an appropriate way to describe the route.
route_long_name	Required	Included (some records)	Populated for all modes. For rapid transit services with branches (Green Line), both route_long_name and route_short_name are populated: route_short_name contains the branch designation only, and route_long_name identifies both the route and the branch. For rail-based and ferry services, route_long_name identifies a lengthier, commonly-used route identifier, such as "Red Line" or "Providence/Stoughton Line". For bus services only, route_long_name provides the typical end points for the route, and should not be used as a standalone route identifier.
route_desc	Optional	Included	Categorizes a route's level of service. For example, identifies whether bus service runs frequently all day, is aimed at weekday commuters, or supplements service on other routes. Possible Values: - Regional Rail - Rapid Transit - Frequent Bus - Local Bus - Supplemental Bus - Coverage Bus - Commuter Bus - Ferry - Seasonal Ferry - Rail Replacement Bus

Field Name	GTFS spec	Status	Notes
route_fare_class	Experimental	Included	Specifies the fare type of the route, which can differ from the service category. This proposal uses this field instead of using fare_rules.txt and fare_attributes.txt as those files currently do not support the entirety of the MBTA's fare and transfer policies. For public-facing applications, we recommend that use and display of route_fare_class be equally or more prominent than route_desc, as passengers often identify routes by their fares. Possible values for the MBTA implementation: • Local Bus • Inner Express • Outer Express • Rapid Transit • Commuter Rail • Ferry • Free • Special
route_type	Required	Included	Indicates the type of vehicle that operates the route. It is not recommended to use this field's values to categorize MBTA service.
route_url	Optional	Included (some records)	
route_color	Optional	Included (some records)	
route_text_color	Optional	Included (some records)	
route_sort_order	Optional	Included	Integer value that can be used for ordering routes in a way that is ideal for presentation to customers.
line_id	Experimental	Included (some records)	References line_id values from lines.txt . Indicates in which grouping of routes this route belongs, if any. For example, route_id 62 may have a line_id value of line-6276. Note that groupings are subject to change without notice.
listed_route	Experimental	Included (some records)	Indicates whether route should be included in a public-facing list of all routes. This is useful for determining which routes which should not be shown by themselves, but rather part of another route. Most uses of this field should incorporate data from multi_route_trips.txt. The following values are valid: • 0 (or empty): Route should be included in a list of routes. • 1: Route should not be included in a public-facing list of routes. For example, in the future, service for route "450W" trips may appear under a route_id of 450W, but with a listed_route value of 1, and all of the route's trips included in multi_route_trips.txt to be displayed together with route_id 450.
network_id	Optional	Included	For use with fare_leg_rules.txt .

route_patterns.txt

Experimental file used to describe the subsets of a route, representing different possible patterns of where trips may serve. For example, a bus route may have multiple branches, and each branch may be modeled as a separate route pattern per direction. Hierarchically, the route pattern level may be considered to be larger than the trip level and smaller than the route level.

For most MBTA modes, a route pattern will typically represent a unique set of stops that may be served on a route-trip combination. Seasonal schedule changes may result in trips within a route pattern having different routings. In simple changes, such a single bus stop removed or added between one schedule rating and the next (for example, between the Summer and Fall schedules), trips will be maintained on the same route_pattern_id. If the changes are significant, a new route_pattern_id may be introduced.

Field Name	GTFS spec	Status	Notes
route_pattern_id	Experimental	Included	Database-unique identifier for the route pattern. For the MBTA, this will generally be a concatenation including the route_id and direction_id. Values from this field are referenced in trips.txt .
route_id	Experimental	Included	Route IDs for which route patterns are valid are referenced in routes.txt .
direction_id	Experimental	Included	Direction IDs for which route patterns are valid are referenced in trips.txt .
route_pattern_name	Experimental	Included	User-facing description of where trips on the route pattern serve. These names are published in the form Origin - Destination or Origin - Destination via Street or Landmark .

Field Name	GTFS spec	Status	Notes
route_pattern_time_desc	Experimental	Included (some records)	User-facing description of when the route pattern operate. Not all route patterns will include a time description. Several examples of possible values include Early mornings only, Weekday evenings only, Saturdays only, and School days only.
route_pattern_typicality	Experimental	Included	Explains how common the route pattern is. For the MBTA, this is within the context of the entire route. Current valid values are: 0 (or empty): Not defined. 1: Typical. Pattern is common for the route. Most routes will have only one such pattern per direction. A few routes may have more than one, such as the Red Line (with one branch to Ashmont and another to Braintree); routes with more than two are rare. 2: Pattern is a deviation from the regular route. 3: Pattern represents a highly atypical pattern for the route, such as a special routing which only runs a handful of times per day. 4: Diversions from normal service, such as planned detours, bus shuttles, or snow routes. 5: Pattern is considered to be a reference, or canonical, pattern for this route and direction containing all physical stops along it, but is not currently scheduled to operate. Travel times for such trips should not be considered representative of actual service. May be used to outline rail routes containing stations with long-term closures. Branched routes may have more than one such pattern per route/direction.
route_pattern_sort_order	Experimental	Included	The route_pattern_sort_order field can be used to order the route patterns in a way which is ideal for presentation to customers. Values are non-negative integers. Route patterns with smaller route_pattern_sort_order values should be displayed before those with larger route_pattern_sort_order values.
representative_trip_id	Experimental	Included	A trip ID that can be considered a canonical trip for the route pattern. Referring to this value in trips.txt and stop_times.txt , a pattern's canonical set of stops and shape ID and be deduced. Note that in this implementation, if a GTFS feed includes trips from multiple schedule ratings, which may include changes in the pattern's routing, the representative_trip_id will come from the schedule rating in effect at the time of the feed's release.
canonical_route_pattern	Experimental	Included	Indicates whether or not a pattern is considered "canonical". This designation is currently used for rail routes to hold a list of all stops that should be shown on line maps, including those served infrequently or which are temporarily closed. Note that this also means canonical route patterns may not always be in operation; use route_pattern_typicality to determine whether this is the case. 0 or empty: Route does not have any canonical patterns defined. 1: Route pattern should be considered canonical for this route in this direction. If branching regularly occurs, this route/direction may have more than one canonical pattern. 2: Route pattern should be not considered canonical for this route in this direction.

shapes.txt

Field Name	GTFS spec	Status	Notes
shape_id	Required	Included	
shape_pt_lat	Required	Included	
shape_pt_lon	Required	Included	
shape_pt_sequence	Required	Included	
shape_dist_traveled	Optional	Included	Distances reported in meters.

stops.txt

Field Name	GTFS spec	Status	Notes
stop_id	Required	Included (persistent)	Expect field length to not exceed 45 characters.
stop_code	Optional	Included (some records) (persistent)	Field is populated only for subway and bus stops, where it has the same value as stop_id. Not prominently displayed or widely used, but visible on some bus stop signs.
stop_name	Required	Included	For most bus stops, indicates geographic location of stop, with first street listed as the street the bus is traveling on, and second street listed (after "@" or "opp") as the cross street. For stops with a parent_station value, gives the name of the parent station. If more specific information about the stop is desired, place see platform_name (to use in conjunction with stop_name) or stop_desc. Bus stops do not contain "stop" with a few exceptions (namely "Terminal B stop 1" and "Terminal B stop 2"). Subway and commuter rail stations do not contain "station" with a few exceptions (namely "North Station" and "South Station").

Field Name	GTFS spec	Status	Notes
			For example, the platform toward Ashmont and Braintree on the Red Line at Park Street will have a stop_name of Park Street.
stop_desc	Optional	Included (some records)	For stops with a parent_station value, indicates the name of the station and that of the specific boarding location. For example, the platform toward Ashmont and Braintree on the Red Line at Park Street will have a stop_desc of Park Street - Red Line - Ashmont/Braintree.
platform_code	Optional	Included (some records)	Indicates the platform identifier for a platform stop (e.g. G or 3), if one is signed at a station. Many stations will not have platform_codes. Words like "platform" or "track" are not included. For example, the platform toward Ashmont and Braintree on the Red Line at Park Street will have a platform_code of 5.
platform_name	Experimental	Included (some records)	Indicates the platform name as labeled at a station, often indicating the direction or destination of services passing through a platform. Words like "platform" or "track" (or the feed's language-specific equivalent) should not be included. For example, the platform toward Ashmont and Braintree on the Red Line at Park Street will have a platform_name of Ashmont/Braintree.
stop_lat	Required	Included (some records)	Children stops with a parent_station may have different stop_lat value as that of the parent. Generic nodes (having a location_type of 3) may not have a stop_lat.
stop_lon	Required	Included (some records)	Children stops with a parent_station may have different stop_lon value as that of the parent. Generic nodes (having a location_type of 3) may not have a stop_lon.
zone_id	Optional	Included (some records)	Populated with Commuter Rail fare zone information at Commuter Rail stations and stops. Special values may be applied at stations at which multiple fare zones or where special event fares are in effect.
stop_address	Experimental	Included (some records)	Optional field which allows a stop to have included a human-readable address.
stop_url	Optional	Included (some records)	Populated for stops which are parent stations (location_type of 1) or scheduled boarding locations (location_type of 0).
level_id	Optional	Included (some records)	Reference to vertical station level from levels.txt .
location_type	Optional	Included	A value of 1 indicates a parent station complex, whereas 2 designates a station entrance/exit, 0 indicates a distinct boarding location, and 3 indicates a generic node within a station, such as the end of a staircase, elevator, or escalator.
parent_station	Optional	Included (some records)	For stops location within stations, the parent_station's stop_id represents the whole facility and the child stop represents a specific boarding area, entrance, or generic node. All subway, Commuter Rail, and CapeFLYER stops have a parent station, as do some bus and Silver Line facilities, such as Dudley.
stop_timezone	Optional	N/A	
wheelchair_boarding	Optional	Included	Additional guidance for bus stops only: • 0: Minor to moderate accessibility barriers exist at the stop. Bus operator may need to relocate bus for safe boarding and exiting. • 2: Significant accessibility barriers exist at the stop. Customers using wheeled mobility devices may need to board at street level.
municipality	Experimental	Included (some records)	Lists the name of the city or town in which the stop is located.
on_street	Experimental	Included (some records)	Provides the full name of the street along which a stop is located. Newer or recently renamed or relocated streets may not properly appear in on_street. Populated for most, but not all, bus stops. Not currently populated for rail or ferry stops, nor for stops with location_type not equal to 0.
at_street	Experimental	Included (some records)	Provides the full name of a stop's nearest cross street. The nearest cross street can be on the same or opposite side of the street as the stop, and thus will not always be the same street as the one referenced in stop_name. Newer or recently renamed or relocated streets may not properly appear in at_street. Populated for most, but not all, bus stops. Note that at_street will not be populated unless on_street is also populated.
vehicle_type	Experimental	Included (some records)	Part of the Google Transit Extensions to GTFS . Describes the mode of transportation used at the stop. Populated for all stops where location_type equals 0, including those which do not have service scheduled in advance. Value definitions match those for route_type; for example, the stop Alewife-02 has a vehicle_type of 1, which defines it as a subway stop.

stop_areas.txt

Denotes groupings of stops useful in the calculation of fares in [fare_leg_rules.txt](#). Typically used to group fare zones for Commuter Rail and express bus routes.

Field Name	GTFS spec	Status	Notes
area_id	Required	Included	
stop_id	Required	Included	

stop_times.txt

Field Name	GTFS spec	Status	Notes
trip_id	Required	Included	
arrival_time	Required	Included	
departure_time	Required	Included	
stop_id	Required	Included (persistent)	
stop_sequence	Required	Included	
stop_headsign	Optional	Included (some records)	Currently only used on commuter rail.
pickup_type	Optional	Included	On commuter rail, a pickup_type of 3 represents a commuter rail "flag" stop marked with "F" on the schedule. Customer must visible to the approaching train to board. On certain bus routes, a pickup_type of 3 represents continuous pickup: The bus will stop at any safe location between GTFS stops to pick up passengers. Customer must wave at the driver to board.
drop_off_type	Optional	Included	On commuter rail, a drop_off_type of 3 represents a commuter rail "flag" stop marked with "F" on the schedule. Customer must notify the conductor to exit. On certain bus routes, a drop_off_type of 3 represents continuous drop off: The bus will stop at any safe location between GTFS stops to drop off passengers. Customer must notify the driver to exit.
shape_dist_traveled	Optional	N/A	
timepoint	Optional	Included	On Commuter Rail, stops that are not timepoints and are not the last stop are labeled "L" on printed schedules to indicate that the train may leave early. MBTA bus and subway operations do not have many scheduled hold points so very few of their stop times have a timepoint value of 1.
checkpoint_id	Experimental	Included (some records)	Identifier for locations at which on-time performance is measured. This includes all MBTA subway stops and major bus stops. Primarily for internal use. Refer also to checkpoints.txt . A checkpoint_id corresponds to a general geographic location (such as a public square or rail station) and can be shared by several routes at different stop_ids in close proximity.
continuous_pickup	Optional	Included (some records)	A value of 0 indicates that continuous pickup is offered between this stop_time and the next stop_time on the trip. An empty value indicates no continuous pickup is offered.
continuous_drop_off	Optional	Included (some records)	A value of 0 indicates that continuous drop off is offered between this stop_time and the next stop_time on the trip. An empty value indicates no continuous drop off is offered.

timeframes.txt

Recently [introduced to GTFS](#), used to describe fares that can vary based on the time of day, the day of the week, or a particular day in the year. Timeframes can be associated with fare products in [fare_leg_rules.txt](#).

Field Name	GTFS spec	Status	Notes
timeframe_group_id	Required	Included	
start_time	Optional (some records)	Included	
end_time	Optional (some records)	Included	
service_id	Required	Included	

transfers.txt

Field Name	GTFS spec	Status	Notes
from_stop_id	Required	Included	
to_stop_id	Required	Included	
transfer_type	Required	Included	Includes usage of the relatively new 4 value to link trips between which riders are allowed to remain aboard.
min_transfer_time	Optional	Included (some records)	As specified in the GTFS standard, this field is the sum of min_walk_time and suggested_buffer_time.
min_walk_time	Experimental	Included (some)	Minimum time required to travel by foot from from_stop_id to to_stop_id.

Field Name	GTFS spec	Status	Notes
		records)	
min_wheelchair_time	Experimental	Included (some records)	Minimum time required to travel by wheelchair from <code>from_stop_id</code> to <code>to_stop_id</code> . If the transfer is not wheelchair accessible, this field will be blank.
suggested_buffer_time	Experimental	Included (some records)	Recommended buffer time to allow to make a successful transfer between two services. This is also partly based on the significance of missing the transfer (due to service frequency).
wheelchair_transfer	Experimental	Included (some records)	Identifies whether a transfer is accessible to customers using a wheelchair. The field can have the following values: 0 (or empty): indicates that there is no accessibility information for the transfer 1: indicates that the transfer is wheelchair accessible 2: transfer not accessible to persons in wheelchairs
from_trip_id	Optional	Included (some records)	If present, specifies that the transfer is exclusively valid from this trip to the trip specified in <code>to_trip_id</code> .
to_trip_id	Optional	Included (some records)	If present, specifies that the transfer is exclusively valid to this trip from the trip specified in <code>from_trip_id</code> .

trips.txt

Field Name	GTFS spec	Status	Notes
route_id	Required	Included (persistent)	
service_id	Required	Included	
trip_id	Required	Included (persistent)	
trip_headsign	Optional	Included	The average <code>trip_headsign</code> length is 13 characters. Expect field length to not exceed 80 characters. Formats: Destination (Example: Park Street) Destination via Street or Landmark (Example: Andrew via South Bay Center) Destination via Street or Landmark & Second Street or Landmark (Example: Wood Island via Market Basket & Admiral's Hill) Destination via Street or Landmark and Second Street or Landmark (Example: Downtown Crossing via Airport and Paradise Road) Destination (Express) (Example: Haymarket (Express)) Destination (Limited Stops) (Example: Boston Medical Center (Limited Stops)) Destination via Street or Landmark (Express) (Example: Woburn via Medford Square (Express)) Destination via Street or Landmark & Second Street or Landmark (Express) (Example: Marblehead via Central Sq & Humphrey St (Express))
trip_short_name	Optional	Included (some records) (persistent)	<code>trip_short_name</code> is only a persistent identifier for commuter rail and CapeFlyer, field is blank for other modes. Even though it is not explicitly an identification field, it can be used to keep track of trips spanning across regular schedule changes. Do not assume that every commuter rail trip will have a <code>trip_short_name</code> .
direction_id	Optional	Included (persistent)	Refer to <code>trip_headsign</code> value to determine which destination indicates inbound or outbound direction for route.
block_id	Optional	Included (some records)	<code>block_id</code> here does not imply that a customer can remain on a vehicle through to the next trip; such instances can instead be inferred by reviewing transfers.txt for transfers with the <code>transfer_type</code> value of 4.
shape_id	Optional	Included	Generally included, but not guaranteed.
wheelchair_accessible	Optional	Included	For MBTA vehicles that are wheelchair accessible, the value is always 1.
trip_route_type	Experimental	Included (some records)	Indicates the type of vehicle that operates the particular trip if the type of vehicle is not the same as specified in the route's <code>route_type</code> (i.e. if part of Blue Line subway service is replaced with a shuttle bus for one weekend). Definitions match those of <code>route_type</code> . For most trips, this is empty.
route_pattern_id	Experimental	Included	Indicates the pattern of a route that a particular trip will operate. Values linked to those in route_patterns.txt .
bikes_allowed	Optional	Included	1 if bikes are allowed on the trip, 2 if they are not. Empty or 0 values make no claim to whether bikes are allowed.

trips_properties.txt

Field Name	GTFS spec	Status	Notes
trip_id	Experimental	Included	Uniquely identifies each trip as defined in trips.txt . Does not need to be unique within this table.
trip_property_id	Experimental	Included	Property for which details are provided in the value field. This may, for example, be a general text note, or designate a trip as being supplemental.
value	Experimental	Included	Value corresponding to a particular property for a trip_id.

trips_properties_definitions.txt

Field Name	GTFS spec	Status	Notes
trip_property_id	Experimental	Included	Trip property ID which is referenced from the property field in trips_properties.txt .
definition	Experimental	Included	Text describing the associated trip_property_id and its intended usage.
possible_values	Experimental	Included	Text describing the allowed values and format of the trip_property_id's respective value field.